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Macro Regime and Threshold Experiments Report

I created an experiment pipeline macro regime v2 experiment suite and spent the last few days reconciling those results against Rohit sir's consolidated plan PDF. I have added answers to the questions that you had asked along with my doubts for you.

Part A: Five regime dimensions

A1: fed_cycle (7 states → 4 states)

Question	Answered?	Answer
Does 4-state collapse give enough observations per state?	Partially	TIGHTENING 763 (40.1%), EASING 727 (38.2%), EASY 384 (20.2%). All exceed ≥30 obs. PIVOTING = 27 (1.4%).
Is any state degenerate?	Mostly yes	No state >80% dominance. PIVOTING statistically thin.
Can we ship v2 fed_cycle to production?	No	Shadow labels exist; production still uses legacy 7-state.

Doubt for Rohit sir: PIVOTING has only 27 Fridays. Merge into EASING, widen the definition, or add PAUSING as a fifth state? Fed is currently on hold with no v2 label (job tracker T-01).

Re Pivoting and easy merge:

1>Note that tightening will also include holding tight.

Agreed. TIGHTENING in the v2 collapse includes both active hiking and holding at plateau rates. Fed on hold at 5.25% for 12 months is still TIGHTENING economically. Legacy HIKING_LATE maps to v2 TIGHTENING in the 4-state collapse.

2> While pivot in language means alter direction i e fed pivots from tightening-to-hold or hold-to-easing or easing to start-to-tighten..... For some reason you are saying pivot shd be merged into easing? SO yes you can widen ranges/reduce regions but be clear on this.

You are right, and I withdraw my earlier suggestion to merge PIVOTING into EASING. Pivot means alter direction (tightening-to-hold, hold-to-easing, easing-to-tighten). The first-cut week after a long tightening cycle has a distinct forward-return profile from month-8 easing. The n=27 PIVOTING bucket is a v2 collapse artefact and I would keep it .

v2 collapsed state	n Fridays	% of sample
TIGHTENING	763	40.1%
EASING	727	38.2%
EASY	384	20.2%
PIVOTING	27	1.4%

I implement a two-tier approach:

1. **Storage / classifier output:** 9 liquidity states permanently (`{LEVEL}_{DIRECTION}`).
2. **Combo hit-rate analytics:** collapse to 4 pure 2×2 buckets when slicing performance (9-way event slices are too thin).

Collapse rules :

9-state label	Collapsed 4-state bucket
EASY_IMPROVING	EASY + IMPROVING
EASY_TIGHTENING	EASY + TIGHTENING
EASY_FLAT	EASY + IMPROVING if prior 4wk WALCL trend positive, else EASY + TIGHTENING
NEUTRAL_IMPROVING	EASY + IMPROVING (NFCI < 0 → lean easy)
NEUTRAL_TIGHTENING	TIGHT + TIGHTENING if NFCI > 0, else EASY + TIGHTENING
NEUTRAL_FLAT	Split by NFCI sign: NFCI < 0 → EASY+FLAT; NFCI > 0 → TIGHT+FLAT
TIGHT_IMPROVING	TIGHT + IMPROVING
TIGHT_TIGHTENING	TIGHT + TIGHTENING
TIGHT_FLAT	TIGHT + dominant 4wk WALCL trend

I prefer keeping NEUTRAL as a separate level in storage (your intuitive preference). For analytics collapse only, NEUTRAL folds by NFCI sign as above.

A2: curve_regime (4 states + fiscal caveat)

Question	Answered?	Answer
Does fiscal deficit >5% weaken inversion signal?	Directionally yes	13 inverted episodes. No-offset (deficit ≤5%): n=12, 41.7% bearish 3m. Fiscal-offset: n=1, 0% bearish (likely 2022-23).
Are 4 curve states stable in backfill?	Yes	All four appear. Oct 2022: 14 inverted weeks via F2 rule.

Doubt for Rohit sir: Fiscal-offset bucket has n=1. Cannot confirm 2022-23 tagging statistically.

Noted and fine to go ahead .

A3: val_regime + CAPE velocity

Question	Answered?	Answer
Does CAPE velocity add signal beyond static level?	Not yet	Level extreme 3m: n=863, 74.2% SPX up, avg +3.08%. Velocity rank delta 6m: n=531, 74.6% up, avg +2.68%. Hit rates differ 0.4 pp only.
Does triple storage help Combo E?	Partially	Combo E high-CAPE 6m: n=507, 79.1% hit, avg +6.41%. Moderate-CAPE bucket: n=0.

Doubt for Rohit sir: Fresh cross into EXTREME CAPE: n=0 with current definition. Cannot test "fresh crossing vs sitting at extreme for 3 years" yet.

Please confirm you have evaluated triple storage of cape= 📊 **Full-history expanding percentile rank, 3-year rolling percentile rank and 8-week rate-of-change (velocity)**

Confirmed. I store all three CAPE representations daily:

Storage type	Definition	Used for
(1) Full-history expanding pctile	CAPE rank vs all history from 1881	Combo detection (unconditional)
(2) 3-year rolling pctile	CAPE rank vs trailing 3Y window	Conviction modifier
(3) 8-week velocity	Rank delta (8wk ROC of percentile)	Fresh-crossing detection

As discussed, take 10 year and

5 year distributions and evaluate impact of triple storage on COMBO E. Please share this excel. The idea is for you to define what is moderate – not say n=0. You define the threshold between moderate and extreme cape and is there a threshold at which value can be added? That is really what we want the macro intelligence agent to do...

I defined moderate CAPE as CAPE 25–35 based on 5Y/10Y distribution percentiles, vs Extreme >35. CAPE has been above 30 continuously since 2018, so the "moderate" bucket has zero recent Combo E fires (all moderate fires are pre-2018).

CAPE distribution table :

Window	Median CAPE	75th pctile	90th pctile
10-year	~28	~32	~35
5-year	~32	~35	~38

Combo E by CAPE bucket at validated horizons (T4 query, n=507 total):

CAPE Bucket	n	Up% 6m	Avg 6m%	PW 6m%	Up% 12m	Avg 12m%	PW 12m%
LOW (<25)	40	100.0%	+9.41%	+9.41%	90.0%	+12.16%	+12.17%
MODERATE (25–30)	127	85.8%	+7.22%	+7.22%	88.2%	+14.29%	+14.30%
HIGH (30–35)	175	77.1%	+6.97%	+6.96%	85.7%	+13.21%	+13.21%
EXTREME (>35)	165	70.9%	+4.46%	+4.46%	64.8%	+5.62%	+5.61%

Finding: MODERATE CAPE Combo E is the strongest bucket historically (85.8% up 6m), but unavailable in modern data. EXTREME CAPE (>35) shows weakest performance, supporting CAPE-conditional Combo E sizing. Level beats velocity by +0.40pp avg return in preliminary test; velocity does not add clear incremental signal.

A4: geo_overlay (6 → 3 states)

Question	Answered?	Answer
Is 3-state geo more reproducible than 6-state?	Yes (qualitatively)	NEUTRAL 1,855 (97.6%), ELEVATED_RISK 25 (1.3%), CRISIS 21 (1.1%).
Does geo slice combo performance meaningfully?	No	FM geo slices mostly n<10. CRISIS n=2. ELEVATED_RISK n=1 at extreme-short FM.

1. Fine lets go with 2 state geo then

I recommend collapsing to 2-state geo: NEUTRAL / ELEVATED going

forward. Research on how large macro funds handle this:

Fund / framework	Approach
Bridgewater	Binary risk-on/off geopolitical overlay: is an active geopolitical event affecting capital flows?
Druckenmiller	Binary "geopolitical tail risk present / absent" rather than categorical taxonomy
Soros reflexivity lens	Focus on whether geo shock is changing market participant behavior, not severity grading

- a. **Prompt flow would be something like this** , (would make it more detailed once you approve , also please do suggest if you want it to specifically and necessarily include anything)
 “Is there an active geopolitical event currently impacting capital flows, commodity prices, or safe-haven demand? Classify as ELEVATED if yes, NEUTRAL if no. Do not use CRISIS as a separate category. Severity is captured by the combo engine through commodity and spread variables.”
- b. Production geo classifier still uses 3-state (NEUTRAL/ELEVATED_RISK/CRISIS). Would update the main code once you approve this approach.

2. Geo slice combo performance

- a. I tested all named combos (A–G) on non-neutral geo dates. Below is the full data

Geo state	Combo	n dates	SPX up% 3m	Avg SPX 3m%
CRISIS	A	1	100.0%	+27.77%
CRISIS	D	1	0.0%	–15.28%
CRISIS	E	7	57.1%	–0.58%
CRISIS	F	7	57.1%	+0.11%

b. Episodes context :

- CRISIS geo = Feb–Jun 2020 (COVID): Combo E/F fired during crash weeks (–11% to –15% 3m); Combo A fired at Apr 2020 bottom (+27.77%).
 - ELEVATED_RISK = Feb–Apr 2022 (Ukraine): Combo E/F mostly negative 3m (–5% to –17%).
 - ELEVATED_RISK = Feb–Apr 2025 (tariff shock): Combo E weak early (–6%), then recovered (+22% by Apr 4).

Geo does not uniformly suppress or boost combos. At $n < 10$ per cell it cannot serve as a reliable standalone filter, but the row-level data is informative for scenario analysis.

A5: liquidity (2 → 4/9 states)

Rohit sir's plan moves liquidity from binary GLOBAL_EASY / GLOBAL_TIGHT to a 2×2 grid: easy vs tight level (from NFCI) crossed with improving vs tightening direction (from WALCL month-over-month change). That is the right design. What the shadow backfill showed is that real data does not always land in one of four clean quadrants.

How the 9 states are built (regime_v2_shadow.py):

Input	Rule	Output
NFCI < -0.3	Easy financial conditions	Level = EASY
NFCI ≥ +0.3	Tight financial conditions	Level = TIGHT
Between -0.3 and +0.3	Neither extreme	Level = NEUTRAL (not in plan's 2×2)
WALCL MOM > +0.3%	Balance sheet expanding	Direction = IMPROVING
WALCL MOM < -0.3%	Balance sheet shrinking (QT)	Direction = TIGHTENING
Between -0.3% and +0.3%	No clear move	Direction = FLAT (not in plan's 2×2)

Label format is {LEVEL}_{DIRECTION}, giving 3 levels × 3 directions = 9 states. The plan's pure 4-state grid would require every Friday to be EASY or TIGHT (no NEUTRAL) and IMPROVING or TIGHTENING (no FLAT).

Full backfill distribution (1,901 Fridays):

State	Count	% of sample	Passes ≥30 obs?
EASY_FLAT	746	39.2%	Yes
EASY_IMPROVING	403	21.2%	Yes
EASY_TIGHTENING	287	15.1%	Yes
NEUTRAL_FLAT	219	11.5%	Yes

State	Count	% of sample	Passes ≥30 obs?
NEUTRAL_TIGHTENING	72	3.8%	Yes
NEUTRAL_IMPROVING	62	3.3%	Yes
TIGHT_IMPROVING	50	2.6%	Yes
TIGHT_TIGHTENING	32	1.7%	Yes
TIGHT_FLAT	30	1.6%	Yes

Worth noting: 50.8% of all Fridays sit in a FLAT direction bucket (965/1,901), and 26.4% are NEUTRAL * level (501/1,901). That is why the backfill produced 9 labels, not 4. Forcing a pure 2×2 would misclassify roughly half the history whenever WALCL MOM is near zero (common during QT pause, post-QE plateau, or noisy weekly prints).

Does WALCL direction add signal? I sliced FM positioning events by liquidity_v2 at the 3m horizon:

[liquidity_v2](#) is the shadow backfill label from [regime_v2_shadow.py](#). It replaces the old binary GLOBAL_EASY / GLOBAL_TIGHT with a 3×3 grid:

Band	Liquidity slice	n	SPX up 3m	Notes
Extreme short FM (<15th)	EASY_FLAT	6	50.0%	No clear edge
Extreme short FM	EASY_IMPROVING	10	60.0%	Similar to FLAT
Extreme short FM	EASY_TIGHTENING	10	50.0%	Similar to FLAT
Extreme short FM	NEUTRAL_*	3 each	33-100 %	Too few to trust
Moderate FM (25th-75th)	EASY_IMPROVING	30	83.3%	Highest slice
Moderate FM	EASY_FLAT	20	70.0%	

Band	Liquidity slice	n	SPX up 3m	Notes
Moderate FM	EASY_TIGHTENING	23	65.2%	~18 pp below IMPROVING
Moderate FM	TIGHT_*	1-2	n/a	Unusable

Please share SPX tables for SPX at 1m, 3m, 6m, 9m, 12m for each Band.

Liquidity State	n fires	Up% 1m	Avg 1m%	Up% 3m	Avg 3m%	Up% 6m	Avg 6m%	Up% 9m	Avg 9m%	Up% 12m	Avg 12m%
EASY_FLAT	1,884	66.9%	+0.30%	76.1%	+2.16%	76.9%	+4.31%	75.3%	+6.51%	70.1%	+8.05%
EASY_IMPROVING	4,215	68.8%	+1.46%	75.2%	+3.16%	70.7%	+4.88%	73.5%	+6.73%	71.7%	+8.37%
EASY_TIGHTENING	4,150	72.1%	+1.60%	81.4%	+4.44%	86.0%	+9.59%	94.1%	+13.70%	94.9%	+17.18%
NEUTRAL_FLAT	426	29.1%	-1.49%	62.9%	+1.16%	70.9%	+5.75%	81.7%	+9.55%	96.7%	+14.59%
NEUTRAL_IMPROVING	720	88.9%	+3.75%	91.3%	+9.80%	85.3%	+12.48%	93.3%	+20.85%	95.0%	+28.23%

NEUTRAL_TIGHTENING	1,941	79.3%	+2.04%	68.6%	+3.41%	93.7%	+8.32%	98.3%	+15.56%	98.8%	+20.21%
TIGHT_FLAT	52	40.4%	-1.61%	48.1%	-8.52%	34.6%	-10.26%	34.6%	-14.60%	34.6%	-16.65%
TIGHT_IMPROVING	1,137	55.9%	-1.03%	30.6%	-5.17%	51.0%	+1.03%	71.8%	+7.72%	71.8%	+15.55%
TIGHT_TIGHTENING	450	45.8%	-2.01%	75.3%	+6.01%	82.4%	+11.87%	82.4%	+17.70%	82.4%	+17.46%

Direction is encoded correctly in the labels (EASY_TIGHTENING vs EASY_IMPROVING are distinct periods). But at the FM-event level, hit rates within the EASY level cluster around 50-60% for extreme short and 65-83% for moderate. The spread is not large enough yet to treat liquidity direction as a standalone combo filter. TIGHT_* buckets are too thin (n=30-50 in the full backfill, n=1-2 at FM events) for any regime-conditional conclusion.

1. **Spread** : You are correct. My prior wording was imprecise. I should have said "range of hit rates" not "spread." What I meant: across EASY sub-states at the FM-event level, hit rates span 50%–83% (extreme short: 50–60%; moderate: 65–83%). The 18 percentage-point gap between EASY_IMPROVING (83.3%) and EASY_TIGHTENING (65.2%) at moderate FM is notable, but sample sizes (n=20–30) leave wide confidence intervals. I have restated this in the main report.
2. **Thin** : Agreed. I show every TIGHT_* observation below so you can have a look at whether 3–4 consistent outcomes at extremes are useful.

Question	Answered?	Answer
Does WALCL direction distinguish tightening vs improving?	Built yes; signal unproven	Labels separate IMPROVING/TIGHTENING/FLAT using WALCL MOM $\pm 0.3\%$ thresholds. Distribution shows direction matters descriptively (EASY_IMPROVING 403 vs EASY_TIGHTENING 287 Fridays). FM slices do not show a reliable performance gap at event level.
Is 4-state 2x2 enough or do we need FLAT variants?	Yes: keep 9 for labels, collapse to 4 for analytics	See recommendation below.

A6 c. descriptively :

A6 d. Do not show a performance gap :

My answer on 4 vs 9 states (from backfill + logic): I recommend a two-tier approach, not picking one number for everything:

1. Production regime storage and classifier output: use 9 states. Forcing 4 would mislabel ~50% of Fridays where WALCL direction is flat or NFCI is in the neutral band. The 9-state scheme is honest to the data and matches how liquidity_v2() is already implemented. Every cell passes the ≥ 30 obs rule in the full backfill (thinnest is TIGHT_FLAT at $n=30$).
2. Combo hit-rate tables, beta filter, and briefing footnotes: collapse to 4 pure 2x2 buckets when slicing performance, because 9-way slices are too thin at the event level (FM extreme short: $n=6-10$ per EASY cell; TIGHT cells nearly empty).

Collapse rules I would use for analytics:

9-state label	Collapsed 4-state bucket
EASY_IMPROVING	EASY + IMPROVING
EASY_TIGHTENING	EASY + TIGHTENING
EASY_FLAT	EASY + IMPROVING if prior 4wk WALCL trend positive, else EASY + TIGHTENING (or hold FLAT as "no direction call")
NEUTRAL_IMPROVING	EASY + IMPROVING (NFCI < 0 -lean easy)
NEUTRAL_TIGHTENING	EASY + TIGHTENING (or TIGHT + TIGHTENING if NFCI > 0)
NEUTRAL_FLAT	NEUTRAL level: split by NFCI sign or exclude from 4-way slice
TIGHT_IMPROVING	TIGHT + IMPROVING
TIGHT_TIGHTENING	TIGHT_TIGHTENING
TIGHT_FLAT	TIGHT+ dominant recent WALCL trend

Do not drop FLAT or NEUTRAL from storage. FLAT is economically real (QT on hold, balance sheet plateau, weekly WALCL noise). NEUTRAL NFCI is real (mildly loose conditions that do not clear the ± 0.3 easy/tight gate). Dropping them would recreate the old binary GLOBAL_EASY/TIGHT problem under a new name.

Remaining doubt for Rohit sir:

- What is the final decision on the 4 vs 9 states?
- The collapse rules for NEUTRAL_FLAT and EASY_FLAT are judgment calls. Do you prefer NEUTRAL level folded into EASY (majority of NEUTRAL Fridays have NFCI slightly negative) or kept as a third level in the classifier prompt only?

Part B: 14th variable and history windows

B1: TWY_ROC (#14)

Question	Answered?	Answer
Did TWY_ROC call Apr 2025 bottom before lagging fed labels?	Yes	Apr 7 2025: TWY_ROC -0.55pp DOVISH (DGS2 3.73%). Legacy fed still TIGHTENING/PAUSING.
Are ± 0.30 pp bands validated?	Partially	Anchor passes (well below -0.30). No full historical band sweep.
Is TWY_ROC excluded from combos?	Yes	298 signatures from 13 vars only. 13,089 generic fires without TWY_ROC leg.

B2: Dual percentile storage

Question	Answered?	Answer
Does dual percentile storage work?	Yes	14,457 rows with both unconditional + regime pctile. 0 unconditional-only rows.
Does <50 fallback logic work when needed?	Not tested	0 fallbacks in backfill. Every fed_cycle had ≥ 50 obs in practice.
Are history windows correct per variable?	No	4 FAIL: HY/VIX/VXTS configured full (plan wants rolling_3y); WALCL was rolling_3y (plan wants full). WALCL fixed in production nightly 2026-06-09 but B4 audit not re-run.

B3: Triple CAPE storage

Question	Answered?	Answer
Which CAPE storage combo predicts best?	Preliminary: level	Level wins avg return by +0.40pp. Not a rigorous multivariate test.
Does velocity beat level for Combo E?	No clear win	High-CAPE Combo E 6m strong regardless of velocity tier.

Deliverable B summary: TWY anchor PASS. Dual storage PASS. Window audit FAIL (4 vars). TWY not in nightly pull yet.

Part C: Emission probability vectors

Question	Answered?	Answer
Can we store 14 daily percentile vectors?	Yes	8,805 daily rows backfilled.
Do sub-threshold readings accumulate useful signal?	Maybe	VIX 65th-79th pctl (below RARE): n=7, 85.7% positive 3m. Too small for statistical gate.
Do vectors detect shifts earlier than binary?	No (so far)	864 VIX RARE events: median lag binary vs vector = 0.0 days.
When can HMM training start?	Not yet	Need 6+ months live vectors. Clock at 0 months.

Doubt for Rohit sir: Prototype HMM did not improve Combo B (-1.2 pp) or D (-1.9 pp). Is ~Dec 2026 still the right HMM target?

Part D: HMM layer

Question	Answered?	Answer
Can HMM posteriors feed classifier as soft prior?	Not in production	Prototype: Jun 2026 sample Risk-On 42.9%, Risk-Off 23.8%, Transition 33.3%.
Does HMM detect shifts before binary thresholds?	Not validated	D3 shift-timing backtest not completed.
HSMM for combo duration exit probability?	Not started	Correctly deferred.
Does HMM improve Sharpe/win rate/drawdown?	No on win rate	with Risk-Off filter (n=56): Combo B: 79.8% 78.6%; Combo D: 28.1% 26.2% (n=103). Sharpe/drawdown not reported.

Question	Answered?	Answer
When is production HMM ready?	~Dec 2026 earliest	Live C1 not wired.

Part E: Cancel probability

Question	Answered?	Answer
Can we compute Combo C cancel prob via MC?	Yes	10,000 GBM paths. WTI leg 8.31%, CPI leg 27.04%, combined 2.25%.
Is probability calibrated to history?	No	4 historical episodes, 0 realized cancels vs 2.25% predicted.
Live cancel % on briefing?	Not yet	Function built, not displayed.
Combo D/F/G cancel formulas?	Documented only	Only Combo C implemented as reusable function.

Part F: Formal regime definitions

F1: TIGHTENING-LATE

Question	Answered?	Answer
Does F1 match Oct 2022?	No	tightening_late_f1 = False. FFR ~3.25% below >3.5% threshold. Legacy was HIKING_LATE. hiking_period_f3 = true.
All fed states have numeric rules?	No	F1 proposed for TIGHTENING-LATE only. Full 4-state quant defs not validated.

F2: INVERTED

Question	Answered?	Answer
Reproducible INVERTED from T10Y2Y?	Yes (shadow)	T10Y2Y < 0 for ≥4 consecutive weeks. Oct 2022: 14 inverted weeks.

F2a: STEEPENING

Question	Answered?	Answer
STEEPENING detectable from numeric rules?	Yes (shadow)	$\geq +15$ bps/4wk RARE, $\geq +40$ EXTREME.

F4: Steepening-of-inversion short grid

Trough	Steepen 4wk	SPX down 3m	n
-50 bps	+15 bps	17.6%	17
-50 bps	+40 bps	25.0%	4
-80 bps	+15 bps	33.3%	9
-80 bps	+40 bps	0%	2

Question	Answered?	Answer
-50 vs -80 trough?	Grid run, no stat winner	-80/+15 best cell (33.3%, n=9) but below 55% bar.
+15 vs +40 steepening?	+15 wins on n	+40 cells n=2-4, meaningless.
Can F4 be promoted on backtest alone?	No	Best 33.3% vs $\geq 80\%$ naming gate. Mechanism+analog only.

Part G: Persistence signals

G1: SEVEN_WEEK_GRIND

Question	Answered?	Answer
Is 7-week grind a standalone short?	No	n=2 episodes. Both negative 6m (avg -5.91%). standalone_short_ok = false. Matches plan (Combo E amplifier).
Wire as Combo E amplifier?	Built, not live	persistence_fires table in shadow. Briefing does not show grind status.

G2: VIX_SUPPRESSED

Question	Answered?	Answer
Does VIX suppressed precede vol spike ~50%?	No	n=1,973 periods. Lead rate to VIX>25 within 35d: 8.5% (168/1,973). Plan claimed ~50%.
Precursor for Combo D?	Directionally yes, weakly	~91.5% false watches if treated as sell trigger. Correct framing: watch flag only.

Part H: Nine-step combo discovery pipeline

Step	Question	Answered?	Answer
1 Detection	298 combos scanned?	Yes	225 with ≥ 1 fire. 13,089 total generic fires.
2 Forward returns	SPX 1m-12m stored?	Yes	Complete in pipeline JSON.
3 Regime tagging	v2 labels used?	Partially	regime tags on existing combo_fires.
4 Surfacing	≥ 3 fires, $\geq 60\%$ HR?	Yes	187 surfaced.
5 Beta filter	Beat hostile regimes?	Yes	132 pass ($\geq 55\%$ and $\geq 60\%$ both reported).
6 Directionality	≥ 2 of 5 dimensions?	Yes	132 pass (same set as beta).
7 Tavila story	Economic narrative?	No	use_claude=False. Skipped.
8 Naming gate	≥ 5 fires, $\geq 80\%$ HR?	Partially	62 promotion candidates. 0 promoted to new named combos.
9 Output table	Cancel prob attached?	No	Part E not wired to live output per combo.

Pipeline funnel:

Stage	Count
Signatures	298
With fires	225
Surfaced	187
Beta + directionality pass	132
Promotion candidates	62
Promoted to production names	0

Doubts for Rohit sir: Beta 55% or 60% for 62 candidates? Re-tag with v2 regimes before final review? Run Tavily step 7 before any promotion?

Part I: Sample-size (covered in brief)

Rule	Threshold	Observed in run
Regime-conditional percentile minimum	≥ 30 obs	PIVOTING fails (n=27)
Fallback to unconditional	< 50 obs	0 fallbacks triggered
Statistical gate	≥ 5 fires	Applied to FM bands, unnamed combos
Mechanism gate	2-4 fires OK	F4, Combo B washout

Two evidence standards are working as designed: F4 correctly stays on mechanism+analog because win rates are too low for statistical promotion.

FM and regime isolation (Rohit sir's additional ask on WhatsApp)

Extreme short FM (<15th pctlile)

Test	n	SPX up 3m	FM wrong/contrary
Combo B fires (incl. WATCH)	89	79.8%	Validates direction vs Rohit sir's 87.5% (7/8)
Raw FM <15 crossings	35	60.0%	Weaker than headline claim

Test	n	SPX up 3m	FM wrong/contrary
EASY fed_cycle slice	6	83.3%	Strongest regime
FLAT curve slice	6	33.3%	Signal breaks here

Question	Answered?	Answer
Is extreme-short FM a contrary indicator?	Conditionally yes	Works best with full Combo B legs, not FM alone.
Why 89 vs "8 confirmed"?	Explained	DB counts WATCH rows with partial legs. Confirmed-only slice not yet run (open PO task).

Extreme long FM (>85th pctile)

Horizon	Raw FM: SPX down	Combo D: wrong	Raw FM: wrong (SPX up)
1 week	59.0%	61.5%	41.0%
3 months	17.9%	82.1%	71.9%

Question	Answered?	Answer
FM wrong 72-85% at short horizons?	Partially	Backtest shows ~59-62%, roughly 10-20 pp below Rohit sir's band.
Signal degrades at 3m?	Yes (correction doesn't happen)	82% FM wrong at 3m for raw band.
Regime impact on Combo D?	Yes	HIKING_LATE 18.3% SPX down 3m (n=197) vs CUTTING_LATE 43.2% (n=155).

Moderate FM (25th-75th)

Metric	Value
Crossings	84
SPX up 3m	76.2%

Metric	Value
Avg 3m return	+3.15%

Rohit sir was right to be skeptical. This looks like equity drift, not an independent FM edge. No alpha from fading or following moderate FM.

Named combos A-G by fed cycle (3m)

Combo	Dir	n	Overall hit	Worst slice	Best slice
A	Bear	174	23% down	CUTTING_LATE 50% (n=26)	QE 20% (n=112)
B	Bull	89	79.8% up	CUTTING_LATE 76% (n=41)	HIKING_LATE 83% (n=48)
C	Bull	4	0% up	n/a	n too small
D	Bear	452	28% down	HIKING_LATE 18%	CUTTING_LATE 43%
E	Bear	507	20% down	CUTTING_LATE ~14-27%	82% (n=212)
F	Bull	704	74.9% up	QE 14%	Flat QE
G		0	fires	CUTTING_LATE 64%	n/a

Question	Answered?	Answer
Does regime materially change combo performance?	Yes	D: 2.4x spread HIKING vs CUTTING. F: 18 pp QE vs CUTTING spread.
Full 5-dimension slicing done?	Partially	fed_cycle reliable; other dims mostly n<10.
Combo C working?	No in sample	4 fires, 0% 3m up, avg +17.8% (market rose against signal).

My doubts and questions

- PIVOTING at n=27: is it a real state or label noise? Merging into EASING is pragmatic but might hide genuine pivot weeks.
- I have not stripped unconditional market drift from the moderate FM 76.2% result. Does it still look special vs buy-and-hold?
- Combo C has n=4 and 0% hit. Should we show C hit rates in the briefing at all until we have more completed episodes?
- VIX suppressed lead rate is 8.5% vs plan's ~50%. Did we measure the wrong window, or is the plan figure an informal estimate that needs updating?
- For production GO-live, what is the rollback plan if v2 fed_cycle labels confuse readers who have used legacy names for months?
- SSI short gate is weak at 26% SPX-down when $SSI \geq 0.85$. Is that acceptable for live sizing, or do we need a tighter gate despite the backtest

UNSENT EMAIL REFERRED TO EARLIER

Divyanshu,

Several architectural clarifications and action items following our review of the macro regime model. Please read this carefully before the next development sprint.

1. HISTORY WINDOWS — CONFIRM THE CORRECT SPEC

Please confirm and implement the following window rules (these supersede any earlier drafts that said 3-year rolling for everything):

- All structural/level variables (CAPE, VIX, yield curve, NFCI, GSR): use FULL EXPANDING HISTORY from variable inception. Never rolling. You want 1970s inversion episodes in the yield curve distribution; you want 1881 CAPE data in the valuation distribution.

- Inflation/rate-of-change variables (WTI 4wk%, CNH 4wk%, WALCL MoM%, CPI surprise): use 3-year rolling. These are regime-dependent and lose meaning over long windows.

- New variable (if we add it): 2-year USD yield 8-week rate of change — 3-year rolling.
- Store both unconditional_pctile (full history) and regime_pctile (conditioned on fed_cycle) for each variable each day. Combo detection uses unconditional. Conviction modifier uses regime_pctile. Fallback to unconditional if regime-conditioned subset has fewer than 50 observations.

2. WHAT "PERCENTILE RANK" MEANS AND WHY IT MATTERS

For each of the 13-14 Runic variables on any given day, you compute one number: where does today's reading sit within the full available history of that variable? VIX at 35 against a 1990-present history = approximately the 92nd percentile = 0.92. This is the percentile rank.

This number is used in two ways:

(a) Combo detection: is it above the RARE threshold (80th-95th percentile depending on variable)?

(b) HMM emission probability: the 13 percentile ranks become the observation vector fed into the HMM each day. The HMM then asks: given a vector like [VIX: 0.92, HY: 0.85, WTI: 0.70, CFTC: 0.12 ...], what is the posterior probability of being in each regime state?

The key difference from what we have now: currently, a variable below its threshold contributes NOTHING to the model. With emission probabilities, a VIX at 0.72 (below the 0.80 RARE threshold) still provides partial evidence for the Stressed state. Sub-threshold readings accumulate evidence week by week. The model becomes sensitive to regime buildup before any single variable crosses its threshold.

Implementation note: you do not need to build the full HMM in the first pass. Start by storing the 13 percentile vectors daily alongside combo_fires. The HMM training can come later once you have 6+ months of clean vectors.

3. FEEDING POSTERIOR PROBABILITIES INTO THE REGIME CLASSIFIER PROMPT

Once the HMM is trained (even a simple 3-state: Risk-On / Risk-Off / Transition), you will have a daily posterior vector, e.g. [Risk-On: 0.65, Risk-Off: 0.25, Transition: 0.10].

This gets added to the Section 5.2 prompt as an additional input:

"HMM posterior state probabilities — Risk-On: 0.65, Risk-Off: 0.25, Transition: 0.10. Use these as a soft prior on regime. If HMM Risk-Off probability is above 0.40, weight toward tighter regime classifications even if individual variables have not crossed thresholds."

This makes the Claude API classifier materially smarter because it receives not just today's individual readings but a model-synthesised view of whether the overall regime pattern matches historical stressed periods.

4. THE GRIND AND SUPPRESSED PERSISTENCE SIGNALS — DEFINE AND IMPLEMENT

The Addendum mentions persistence signals but they are not yet formally defined. Please implement the following two as the first batch:

7WK_GRIND: SPX weekly close > prior week's close by at least +0.5% for 7 consecutive weeks. Saturday job. Check last 7 weekly SPX closes. If condition met, flag streak and store in persistence_fires table with date and streak_length. Wire as AMPLIFIER to Combo E (valuation extreme) — when both fire simultaneously, the structural warning is elevated. Analysis confirms this is not a reliable standalone short signal (average 6m return after grind fires is +7.5%, 100% positive). Use only as amplifier, not trigger.

VIX_SUPPRESSED: VIX close < 15 for 10 consecutive trading days. Daily job. Store in persistence_fires table. Wire as PRECURSOR to Combo D (FOMO top/euphoria). Historical analysis: 12 instances since 1990, approximately half (6/12) were followed by VIX events above 30 within 12 months. The other half saw markets continue without a major shock. This is a WATCH LIST signal, not a sell trigger.

For both: store start_date, most_recent_date, streak_length, and combo_link in persistence_fires. Surface on the nightly briefing with explicit language: "[GRIND/VIX_SUPPRESSED] active for N days/weeks — monitor for Combo D/E confirmation."

5. HIKING_LATE — PLEASE FORMALISE THE QUANTITATIVE DEFINITION

The current classifier prompt infers HIKING_LATE from context. This is not reproducible or testable. Please add explicit quantitative trigger rules to the prompt so the classification can be validated:

HIKING_LATE: FFR > 3.5% AND FFR has risen > 150bps in prior 12 months AND (10Y-2Y < -30bps OR rate of hike deceleration is negative over the last two Fed meetings).

HIKING_EARLY: FFR rising AND total hike cycle < 150bps.

CUTTING_EARLY: FFR falling AND first cut was within last 6 months.

CUTTING_LATE: FFR falling AND has fallen > 150bps from cycle peak.

PAUSING: FFR unchanged for 3+ consecutive meetings.

QE: WALCL rising, FFR at or near zero lower bound.

QT: WALCL falling at > \$50bn/month rate.

October 2022 canonical example: FFR ~3.25%, risen 300bps in 9 months, curve -50bps = HIKING_LATE. March 2020: FFR at zero, WALCL expanding rapidly = QE. Please validate these labels against all historical backfill dates and report any inconsistencies.

6. COMBO DETECTION — CONFIRM WE HAVE SWITCHED TO EXPANDING HISTORY

The spec previously said 3-year rolling percentile for all variables. The updated approach is full expanding history for structural variables (see item 1 above). Please confirm in code that combo detection for CAPE, VIX, NFCI, yield curve, and GSR uses full expanding history, not a 3-year rolling window. Update and re-run the historical backfill if this was not already the case. The change will affect percentile rankings for these variables, which will affect which combo fires are recorded historically.

7. BETA ADJUSTMENT FOR UNNAMED COMBOS — UPDATED DEFINITION

When testing the 291 unnamed combos, the hit rate gate is 60% with minimum 3 fires (for surfacing) and 75% with minimum 5 fires (for naming). To deal with the risk that a combo is lucky due to market beta rather than genuine signal, add a directionality consistency test:

A bullish unnamed combo must show hit rate $\geq 50\%$ in HIKING or INVERTED curve periods (not just in QE/bull periods) to be considered genuine. Define INVERTED period as: 10Y-2Y spread $< -20\text{bps}$ sustained for at least 4 consecutive weeks. Define HIKING period as: FFR rising, total hike cycle $> 100\text{bps}$. A combo that performs well only in easy monetary conditions is picking up beta, not alpha.

The 55% threshold for this test (not 60%) is deliberate — HIKING/INVERTED periods are naturally adverse for longs, so requiring 60% there would eliminate most genuine signals too. 55% says the signal still works in hostile conditions, just modestly. You can test 55% vs 60% and report both.

8. SAMPLE SIZE NOTE FOR REGIME-CONDITIONAL COMBOS

When you compute `regime_pctile` for a variable conditioned on `fed_cycle`, the regime-conditioned subset may have fewer than 50 observations for some variable-regime combinations. In that case, fall back to the unconditional percentile rank and log which was used. Do not make investment decisions based on a regime-conditional percentile rank computed from fewer than 30 observations. The 3-4 instances of a given combo firing in HIKING_LATE since 1990 is too few for statistical confidence alone — which is exactly why we use scenario analysis (does the outcome have an economic story?) alongside the quantitative hit rate.

Rohit

CANCELLATION EXPLAINER - PLEASE ASK ME IF ANY QUESTIONS. I AM HAPPY TO EXPLAIN.

INPUTS — COMBO C CANCEL CONDITION

Current WTI (\$)	\$91	\$91
WTI annualised vol (%)	35	35%
Fridays already cleared	0	0
MC simulations	20k	20k
WTI REFERENCE PRICES (4 WEEKS BEFORE EACH FRIDAY)		
Friday 1 ref — May 9 est (\$)	\$95	\$95
Friday 2 ref — May 16 est (\$)	\$97	\$97
Friday 3 ref — May 23 est (\$)	\$95	\$95
Friday 4 ref — May 30 est (\$)	\$91	\$91

OPTIONS ANALOGY — PER-FRIDAY "STRIKES"

Cancel threshold for each Friday = $WTI_{ref} \times 1.05$. This is the "strike" of the digital put. WTI must close BELOW this level for the cancel condition to be met on that Friday.

Jun 6	ref \$95 → K \$99.75	moneyness -8.8%	cur 4wk -4.2%	97.2%
Jun 13	ref \$97 → K \$101.85	moneyness -10.7%	cur 4wk -6.2%	95.3%
Jun 20	ref \$95 → K \$99.75	moneyness -8.8%	cur 4wk -4.2%	87.2%
Jun 27	ref \$91 → K \$95.55	moneyness -4.8%	cur 4wk 0.0%	70.9%



CANCEL PROBABILITY — OVERALL RESULT

P(combo C cancels — all 4 consecutive Fridays clear)



97%	95%	87%	69.7%
P(Friday 1 clears)	P(Friday 2 clears)	P(Friday 3 clears)	P(all 4 — MC)

Moderate-high cancel probability (69.7%). WTI has meaningful headroom to the strike levels but a $\pm 32\%$ weekly move could breach one of the thresholds. Track WTI weekly. CPI leg needs checking independently.

CANCEL OPTION MATH

Per-Friday digital put probability:

$$d2 = [\ln(S/K) + (-\sigma^2/2) \cdot T] / (\sigma \cdot \sqrt{T})$$

$$P(WTI_Friday < K) = N(-d2)$$

Current inputs: $S = \$91$, $\sigma = 35\%$, $\sigma_weekly = 4.85\%$

Strikes: $K1 = \$99.8$, $K2 = \$101.9$, $K3 = \$99.8$, $K4 = \$95.5$

Moneyness ($S/K-1$): -8.8% , -10.7% , -8.8% , -4.8%

Path dependency: 4 consecutive readings required (not just any 4 of 4).

Overlapping 4wk windows $\rightarrow \rho \approx 0.75$ between adjacent Fridays.

Monte Carlo (GBM, 20k paths) captures correlation exactly.

Joint $P(\text{cancel WTI leg}) = 69.7\%$ [MC estimate, $\pm 0.6\%$ 95% CI]

The cancel condition is a **path-dependent digital barrier** — similar to an Asian digital or a consecutive-touch barrier option. Vanilla $N(d2)$ gives the marginal probability per Friday. The consecutive requirement introduces correlation across the four readings (overlapping 4-week windows share 3 of 4 weeks $\rightarrow \rho \approx 0.75$). Monte Carlo with GBM paths captures this exactly.

The CPI leg is an independent condition. Total $P(\text{combo C cancels fully}) = P(\text{all 4 WTI Fridays clear}) \times P(\text{CPI non-hot for 2 consecutive prints})$. Treat the two legs as independent unless there is a structural link (e.g. oil-driven CPI).

AN EXPLAINER FROM CLAUDE

That's the rigorous version of what you described. A few notes on the framework:

What makes this genuinely options-theoretic and not garbage

Each Friday's cancel probability is computed as $N(-d2)$ — exactly the risk-neutral probability that a log-normally distributed asset ends up below the strike at expiry. The "strike" is $WTI_ref_k \times 1.05$. The "implied forward" is the current WTI price projected by GBM. The "vol" is the annualised WTI volatility (default 35%, calibrate to whatever your

realized vol estimate is). The moneyness display tells you whether today's WTI is already below each strike and by how much.

Why Monte Carlo rather than just multiplying the four marginals

Multiplying the four $N(-d2)$ values would assume independence — i.e. that Friday 1 and Friday 2 are unrelated events. They're not: consecutive 4-week windows share three of four weeks, so adjacent readings have $\rho \approx 0.75$ under GBM. Monte Carlo generates full price paths, so the correlation structure is handled correctly without any approximation. The 95% confidence interval shown in the formula box tells you how precise the Monte Carlo estimate is — at 20k paths it's typically ± 0.5 -1.5%.

Extending this to any Combo cancel condition

The same framework generalises directly. For any condition of the form "variable X stays below/above threshold Y for N consecutive periods":

- S = current value of variable X
- K = threshold Y
- σ = historical volatility of X
- T_k = time horizon for period k
- Correlation between periods = overlap fraction

The CPI leg of Combo C is a separate digital binary — $P(\text{CPI print} \leq \text{consensus for 2 consecutive releases})$. That's not a continuous price process but a discrete event, so it needs a different calibration: historically what fraction of CPI prints have come in $\leq$ consensus? That's your per-print probability, and since consecutive prints are largely independent, $P(2 \text{ consecutive}) \approx p^2$. Multiply that by the WTI leg probability to get total $P(\text{Combo C fully cancels})$.

Why this matters for position sizing beyond Combo C

This same logic applies to any time-based or level-based combo rule. Combo F's 26-week validity window = $P(\text{still active in week } N) = 1 \text{ if } N \leq 26, 0 \text{ if } N > 26$ (deterministic, no probability needed). Combo D's 3-10 day short horizon = the digital put expires fast, so high vol makes it either clear quickly or fail quickly. Combo G's "leads vol spike 3-6 weeks" = you're estimating $P(\text{vol spike materialises within the window})$, which is essentially a variance option. The framework is the same in each case — what varies is the underlying process and the threshold structure.

DOCUMENT SHARED EARLIER CONTAINS REGIME SCORE, FIT CALCULATIONS, ETC

JUNE 04 DOCUMENT –DIVYANSHU FINAL ADDENDUM REGIME SCORE FED CYCLE –

MACRO INTELLIGENCE AGENT — FINAL BUILD SPEC ADDENDUM

Divyanshu | June 4, 2026 | Covers: Fed cycle classification · Regime score methodology · FM covering velocity · Combo predictive power testing · 291+ unnamed combo engine

Divyanshu,

This addendum consolidates all open specification items and introduces the final testing framework for the predictive power of the combo engine. Read alongside the main spec v2.2 and addendum already sent.

SECTION A — FED CYCLE CLASSIFICATION (Python rules, final version)

A1. The AND condition for HIKING_LATE and CUTTING_LATE

Important naming note: HIKING_LATE means LATE IN THE HIKING CYCLE (cycle is mature, >6 months old). It does NOT mean 'late to hike' or any comment on Fed policy timing.

The two conditions must BOTH be true (AND, not OR):

- Time condition: >6 months since first hike of this cycle (HIKING_LATE) or >9 months since first cut of this cycle (CUTTING_LATE)

- **Market confirmation: 2-year UST yield 8-week change < -20bps (LATE in hiking = market pricing end of cycle) or > +20bps (LATE in cutting = market pricing end of cuts)**

Why AND and not OR: Time alone can misclassify early — a 7-month-old hiking cycle where 2Y yields are still rising is not 'late'. Market confirmation via 2Y yield provides the forward-looking layer that the calendar-based rule misses. They are complementary, not redundant.

```
def classify_fed_cycle(fed_funds_rate_series, us2y_series, walcl_series):
    current_rate = fed_funds_rate_series.iloc[-1]
    us2y_change_8wk = (us2y_series.iloc[-1] - us2y_series.iloc[-40]) * 100 # bps
    walcl_mom = (walcl_series.dropna().iloc[-1] / walcl_series.dropna().iloc[-5] - 1) * 100
    if walcl_mom > 0.8: return 'QE'
    if walcl_mom < -0.8: return 'QT'
    # Find first hike/cut of current cycle
    rate_diff = fed_funds_rate_series.diff()
    # HIKING LATE: >6 months AND 2Y already falling >20bps
    if us2y_change_8wk > 20: # market still pricing hikes
        first_hike = get_first_cycle_action(rate_diff, direction='hike')
        months = (fed_funds_rate_series.index[-1] - first_hike).days / 30
        if months > 6 and us2y_change_8wk > 20: return 'HIKING_LATE'
        if months <= 6: return 'HIKING_EARLY'
    # CUTTING LATE: >9 months AND 2Y already rising >20bps
    if us2y_change_8wk < -20: # market still pricing cuts
        first_cut = get_first_cycle_action(rate_diff, direction='cut')
        months = (fed_funds_rate_series.index[-1] - first_cut).days / 30
        if months > 9 and us2y_change_8wk < -20: return 'CUTTING_LATE'
        if months <= 9: return 'CUTTING_EARLY'
    # Pausing — use 2Y direction for sub-classification
    if us2y_change_8wk < -15: return 'PAUSING_DOVISH'
    if us2y_change_8wk > +15: return 'PAUSING_HAWKISH'
    return 'PAUSING_NEUTRAL'
```

A2. What to test / validate empirically

Divyanshu: once the historical database is populated, run the following validation. The goal is to confirm whether the AND condition (time + 2Y yield) improves regime-adjusted hit rates versus the time-only condition.

Test	What to run	Expected finding	Which combos to test on
Test A1: Time-only classification	Classify each historical combo fire using time-based rule only (>6m = HIKING_LATE, >9m = CUTTING_LATE). Compute hit rate for Combo B and Combo F split by regime.	Baseline hit rates: B in HIKING_LATE ~68%, B in CUTTING_LATE ~91%	Combo B, Combo F primarily. Also C, E.
Test A2: AND condition classification	Reclassify using AND condition (time + 2Y yield 8wk change). Some HIKING_LATE dates will reclassify to HIKING_EARLY if 2Y not yet falling. Recompute hit rates.	Hit rate gap between LATE and EARLY should widen if AND condition is better. HIKING_LATE hit rate should increase.	Same combos. Compare directly to Test A1.
Test A3: 2Y yield only (no time condition)	Classify purely by 2Y yield 8wk direction (< -20bps = CUTTING, > +20bps = HIKING, ±20bps = PAUSING). Compute hit rates.	Should produce leading indicators. May show higher predictive power at short horizons (1-3m) but noisier.	All named combos.
Test A4: Union set analysis	Find dates where Time-only fires CUTTING_LATE AND 2Y-yield fires CUTTING simultaneously. Compute hit rate on intersection vs each alone.	Intersection should have highest hit rate — most conservative but most reliable.	Combo B (most important to validate).

SECTION B — REGIME SCORE: PROPER HEURISTIC METHODOLOGY

B1. The problem with the earlier approach

The arbitrary direction weights (± 0.5 , ± 0.8 , ± 0.6 etc.) and probability estimates had no formal basis. They were heuristic guesses. The correct approach uses the combo hit rates directly as weights, applies a time-decay for time-limited signals, and produces a score with a natural interpretation.

B2. Correct formula

Score for each combo = $(\text{hit_rate} - 0.5) \times \text{direction} \times \text{active} \times \text{time_decay}$

Where:

- $(\text{hit_rate} - 0.5)$ = the 'edge above chance'. A 50% hit rate gives 0 edge. A 90% hit rate gives 0.40 edge. A 60% hit rate gives 0.10 edge. This is the only defensible weight — it is derived directly from empirical data.

- direction = +1 for bullish combos (B, F, A-EASY), -1 for bearish combos (C, D, E, G, A-TIGHT)
- active = 1 if all required conditions met (per combo spec), 0 otherwise. If E requires 2 of 3 and 2 are met, active = 1. Not fractional.
- time_decay = for time-limited signals: remaining_window / total_window. Combo F week 10 of 26 = 16/26 = 0.615. For unlimited signals (E, A): time_decay = 1.0. For approaching-cancellation signals (C): time_decay = weeks_remaining_before_likely_cancel / median_duration.

```
# Today June 4, 2026 — computed regime score
combos = {
  'C': {'hit_rate': 0.83, 'direction': -1, 'active': 1,
        'time_decay': 0.25}, # wk12 of 16 MEDIUM, 4wks to cancel window
  'E': {'hit_rate': 0.73, 'direction': -1, 'active': 1,
        'time_decay': 1.00}, # no time limit
  'F': {'hit_rate': 0.85, 'direction': +1, 'active': 1,
        'time_decay': 0.615}, # wk10 of 26
  'B': {'hit_rate': 0.91, 'direction': +1, 'active': 0,
        'time_decay': 1.00}, # CFTC met but VIX and HY not
}
score = sum((v['hit_rate']-0.5) * v['direction'] * v['active'] * v['time_decay']
            for v in combos.values())
# = (0.33*-1*1*0.25) + (0.23*-1*1*1.0) + (0.35*+1*1*0.615) + (0.41*+1*0*1.0)
# = -0.083 + -0.230 + +0.215 + 0.000
# = -0.098 → MILDLY BEARISH (scale approx -1 to +1)
```

Why this is better: Every weight traces directly to an empirical hit rate. Time decay prevents expired signals from dominating. The scale [-1, +1] has natural meaning: 0 = no edge, +1 = all bullish combos firing at maximum hit rate simultaneously.

Important: The direction weights and hit rates will be refined once the historical database has 30+ instances per combo. Until then, use the confirmed hit rates from the spec (B=0.87-0.91, F=0.78-0.85, C=0.83, E=0.73, D=0.72-0.85, G=0.75).

SECTION C — FM SHORT COVERING VELOCITY

C1. What to compute from CFTC historical data

Fast Money (FM) = Leveraged Funds in the CFTC TFF report. Current FM net position as of May 26, 2026: -2,071,353 contracts (5th–8th percentile of 3yr rolling window = extreme short). SPX has rallied ~25% since April 2025 low yet FM has barely covered. This divergence is unusual and important for Combo F.

Divyanshu: pull the full CFTC TFF historical CSV for S&P 500 E-mini futures (2006–2026) from [cftc.gov](https://www.cftc.gov) annual files. Compute the following:

Metric	Method	Output
All weeks FM < 10th pctile (3yr rolling)	Compute FM net = Lev_Long - Lev_Short for each week. Compute 3yr rolling percentile. Find all weeks < 10th pctile.	List of episode start dates
Weeks from trough to 50th pctile (neutral)	For each episode: count weeks from first sub-10th reading until FM crosses 50th pctile. Record how fast covering occurred.	Distribution of covering duration: min, median, max weeks
Weekly covering rate (contracts/week)	(50th pctile level – trough level) + weeks elapsed for each episode	Avg weekly covering rate
SPX return during covering window	SPX close at episode start vs SPX close when FM reached 50th pctile	Confirms whether SPX gain drives covering or covering drives SPX gain
Current anomaly flag	Current episode: 10 weeks elapsed, FM moved from ~2nd to ~7th pctile. Compare velocity to historical avg.	If current velocity < historical avg by >50%: flag as anomaly — covering phase has further to run

Why this matters for Combo F: Combo F's +9.46% average 6m return is driven mechanically by short covering. If FM covering velocity is historically 8 weeks to neutral and we are at week 10 with FM still at 7th pctile, one of two things is true: (a) this episode's covering phase will be unusually prolonged and SPX will continue higher, or (b) the covering has stalled because funds are right to stay short. Distinguishing these two cases requires knowing historical velocity. That is what this analysis provides.

SECTION D — REGIME SCORE VALIDATION TESTS

D1. How to validate the regime score formula

Once the historical database is populated (combo_fires + forward_returns + macro_regime), run:

Test	Method	Success criterion
Regime score vs forward SPX returns	For each historical date in combo_fires, compute regime score using the $(hit_rate - 0.5) \times direction \times active \times time_decay$ formula. Run Spearman correlation between score and SPX 3m/6m forward return.	Spearman $\rho > 0.30$ at 3m, > 0.35 at 6m. p-value < 0.05 .
AND condition improvement test	Recompute regime score using time-only fed cycle vs AND condition. Compare correlations.	AND condition should produce higher ρ at 6m horizon.
Hit rate weight calibration	Test alternative weights: equal weighting (all combos weight 1), hit-rate weighting (current proposal), and optimised weighting (minimise MSE on forward returns). Compare out-of-sample performance on held-out 20% of dates.	Hit-rate weighting should outperform equal weighting. Optimised weighting may overfit — report both.
Time decay sensitivity	Test regime score with $time_decay=1.0$ (no decay) vs proposed decay formula. Compare predictive power.	Decay should improve predictive power at short horizons (3m) by preventing stale signals from dominating.

SECTION E — 291+ UNNAMED COMBO ENGINE: PREDICTIVE POWER FRAMEWORK

E1. The problem: how do we know which unnamed combos are signal vs noise?

The 298-combo engine will discover combos that fire with high hit rates by chance — especially at small N (3-5 instances). We need a rigorous gating framework that distinguishes genuine signal from statistical noise.

E2. The three-gate framework

Gate	Condition	Rationale
------	-----------	-----------

Gate 1: Minimum instances	≥5 historical fires (not 3) before a combo is considered for naming. At N=3 the 80% hit rate threshold (2.4 successes) has a p-value of ~0.10 under the null of 50% hit rate — not significant. At N=5 with 80% hit rate, p-value ~0.03.	Statistical minimum for significance given fat-tailed financial data
Gate 2: Hit rate threshold	≥80% hit rate at 3m for bullish combos, ≥75% at 6m. For bearish combos: ≥75% at 3m. These thresholds ensure the hit rate is meaningfully above the unconditional SPX return frequency (~60% of 3m periods are positive).	Ensures edge above unconditional baseline
Gate 3: Economic mechanism check (Claude API)	Pass every Gate 1+2 combo to Claude API with: 'What causal economic mechanism explains why these two/three variables being simultaneously extreme predicts SPX forward returns?' If Claude cannot articulate a plausible mechanism, flag as STATISTICAL — do not name it.	Prevents spurious correlations from entering the rule library
Gate 4: Out-of-sample validation	Split historical data 80/20. Gate 1-3 applied on training set only. Validate hit rate on held-out 20%. If hit rate drops >15pp out-of-sample, mark combo as OVERFIT — do not use.	Prevents overfitting on small samples

E3. Persistence signals (separate from cross-sectional combos)

The 298-combo engine detects cross-sectional extremes (multiple variables extreme on the same day). Persistence signals detect time-series patterns within a single variable. These are architecturally separate but feed the same output JSON. Build the persistence engine as a separate weekly job:

Signal name	Variable	Condition	Consecutive periods	Gate condition
7WK_GRIND	SPX weekly return	>+0.5%	7 weeks	≥3 historical instances, SPX forward return distribution shows mean reversion signal
3WK_SURGE	SPX weekly return	≥+3%	3 consecutive weeks	2 instances since 1980 — too few for Gate 1. Store as WATCH only, do not promote to named combo.
VIX_SUPPRESSED	VIX daily close	<15	10 trading days	≥5 instances. Test whether VIX suppression >10 days predicts higher vol realization in next 30 days.

HY_GRIND_TIG HT	HY OAS 4wk change	tightening (≥-20bps)	8 consecutive weeks	≥5 instances. Test forward HY spread reversion and SPX return.
OIL_VOLATILE	WTI weekly	±5%+ alternating	4 consecutive weeks	Directional volatility — test whether this precedes trend resolution.

E4. Variable count clarification

The system now has 13 variables (12 original + 2-year UST yield as an additional variable used for fed_cycle classification). The 2Y yield is NOT added to the combinatorial engine as a 13th combo variable — it is used only in the regime classifier. The combo engine remains $12C1+12C2+12C3 = 298$ combinations. If the 2Y yield is later added as a combo variable, the engine becomes $13C1+13C2+13C3 = 13+78+286 = 377$ combinations. That is a v2 decision.

SECTION F — COMBO A NAMING CHANGE

Final naming for Combo A: EASY MONEY (when FCI variables signal loose/expanding conditions) and TIGHT MONEY (when FCI variables signal tight/contracting conditions). Replace all instances of BRAVE/FEARFUL/BULLISH ENVIRONMENT in the code and output JSON.

Old label	New label	JSON field value
Combo A BRAVE	EASY MONEY	combo_a_regime: 'EASY_MONEY'
Combo A FEARFUL	TIGHT MONEY	combo_a_regime: 'TIGHT_MONEY'
NFCI 'BRAVE extreme' zone	EASY EXTREME (NFCI < -0.6)	nfc_i_zone: 'EASY_EXTREME'
NFCI 'RARE LOOSE' zone	EASY (NFCI < -0.3)	nfc_i_zone: 'EASY'
NFCI 'RARE TIGHT' zone	TIGHT (NFCI > +0.3)	nfc_i_zone: 'TIGHT'
NFCI 'EXTREME TIGHT' zone	CREDIT STRESS EXTREME (NFCI > +0.8)	nfc_i_zone: 'STRESS_EXTREME'

Divyanshu — this addendum + the main spec v2.2 + the data source PDF = complete build specification. If anything is unclear, ask before building. The testing framework in Section D and E is particularly important — do not skip it.

Rohit